Ayemere J. Bellowalker

Bioinformatics Scientist

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SUMMARY

Bioinformatics Scientist specialising in scalable data production pipelines and biological data integration. Experienced in designing modular Nextflow workflows for large-scale sequencing datasets (>2.3B reads), with a focus on reproducibility, automation, and efficient data processing across HPC and cloud environments. Strong programming background in Python, R, and Bash, with experience developing pipeline components, debugging workflows, and managing high-throughput data. Particularly interested in biological database systems, data standardisation, and improving the scalability and quality of protein and genomic data resources.

TECHNICAL SKILLS

Programming & Workflow Development:
Python, R, Bash, Java, Nextflow, Git

Bioinformatics & Genomics:
Metagenomics, 16S rRNA, Next-generation sequencing, metagenome-assembled genomes (MAGs)

Tools & Databases:
QIIME2, Kraken2, MEGAHIT, MetaBAT2, SemiBin2, GTDB-Tk, eggNOG, MOTHUR, Silva

Infrastructure & Compute:
AWS EC2, Docker, Linux/Unix, High-performance computing, Remote (SSH)

Data Science:
Statistical modelling, Clustering, Data visualisation, Graph-based modelling, Machine learning

Data Engineering & Databases:
SQL (basic), Data parsing, Data transformation, Structured data formats (CSV, TSV, JSON)

SELECTED PROJECTS (BIOINFORMATICS PIPELINES)

Scalable Metagenomic Analysis Pipeline (Nextflow + Docker + AWS)

GitHub Repository: Scalable-Metagenomic-Analysis-Pipeline

- Built a modular, production-grade pipeline for genome-resolved metagenomics
- Processed >2.3 billion sequencing reads across large-scale datasets
- Reconstructed 1,429 MAGs, including 255 medium/high-quality genomes
- Identified 80 putative novel microbial genomes and 469 nitrogen metabolism genes
- Integrated Kraken2, MEGAHIT, MetaBAT2, SemiBin2, DAS tool, CheckM2, GTDB-Tk, and eggNOG

Amplicon Analysis Pipeline (QIIME2 + R)

GitHub Repository: Amplicon-Analysis-Pipeline

- Built an end-to-end 16S rRNA workflow for microbial community analysis
- Implemented denoising (DADA2), taxonomic classification (SILVA), and diversity analysis
- Performed statistical modelling using phyloseq, DESeq2, and vegan
- Generated publication-ready visualisations of microbial community structure

Biological Data Integration Pipeline

- Designed a Python-based pipeline to integrate and standardise biological annotation data
- Parsed and structured genomic and functional annotation outputs into tabular and relational formats
- Implemented automated data cleaning, validation, and transformation workflows
- Generated queryable datasets for downstream biological analysis and reporting
- Applied modular design principles to enable scalable data processing

PROFESSIONAL EXPERIENCE

Doctoral Researcher October 2022 - Present

Harper Adams University - Telford, UK

- Designed and maintained production-grade bioinformatics pipelines for large-scale sequencing data processing, including debugging, optimisation, and workflow monitoring
- Diagnosed and resolved pipeline execution failures across HPC and cloud environments, improving workflow robustness and reproducibility
- Reconstructed 1,429 MAGs, including 255 medium/high-quality genomes (≥50–90% completeness)
- Implemented cloud-based execution (AWS EC2), optimising compute efficiency for large datasets
- Integrated multi-omics datasets to model microbial metabolic function, with a focus on nitrogen metabolism pathways
- Applied graph-based simulation frameworks (Java/R; microPop, eGut) to model microbiome system dynamics, supporting scenario testing and predictive analysis.
- Delivered decision-ready outputs with cross-functional collaboration in diverse working environments, leveraging strong communication and presentation skills to support team-oriented research

Laboratory Technician July 2022 - September 2022

Intertek Food Services - Stoke-on-Trent, UK

- Performed high-throughput PCR and ELISA assays on 200+ food samples daily, ensuring accurate and timely data generation in a regulated laboratory environment.
- Contributed to microbial and chemical risk assessments through precise sample preparation, assay execution, and ISO-compliant data recording.
- Maintained strict adherence to biosafety, contamination control, and quality management protocols within a GLP-aligned workflow.

EDUCATION

Doctor of Philosophy (Ph.D.) October 2022 - Present

Harper Adams University - Telford, UK

Research: Enhancing the role of the rumen microbiome in improving nitrogen use efficiency in dairy cows

Bachelor of Science (Honours) September 2019 - July 2022

Staffordshire University - Stoke-on-Trent, UK

1st Class Honours - Biological Science with Genomics & Bioinformatics

PUBLICATION & CONFERENCE PRESENTATION

- Manuscript in preparation: “Essential oils and live yeast, when added to diets of lactating dairy cows, alters the abundance of methanogens...” (target submission: Q4 2026)
- Presented at the 14th International Gut Microbiome Symposium, France (June 2025)